

Task 4 – Budget Threshold Alert Tool

Project Summary

The Budget Threshold Alert Tool is an automated monitoring system built with Google Sheets and Google Apps Script. It reads expense data by category, compares actual spending with the allocated budget, detects overspending, and sends a professional email alert whenever a category exceeds its budget threshold.

Task Requirements and Evidence

Requirement	Implementation	Status
Reads a sample expense spreadsheet organized by category	Reads the Expense Data sheet using SpreadsheetApp and processes each category row.	Met
Compares actual spend to a set budget per category	Checks Actual Spend against Budget and calculates amount and percentage over budget.	Met
Sends an alert automatically when a category goes over	Builds an HTML alert email and sends it through GmailApp when overspending is detected.	Met
Deliverable: working script + screenshot of triggered alert	Exact working script and real Gmail alert screenshot are included below.	Met

Triggered Alert – Real Sample Output

The screenshot below shows the actual budget threshold email generated by the system after overspending was detected.

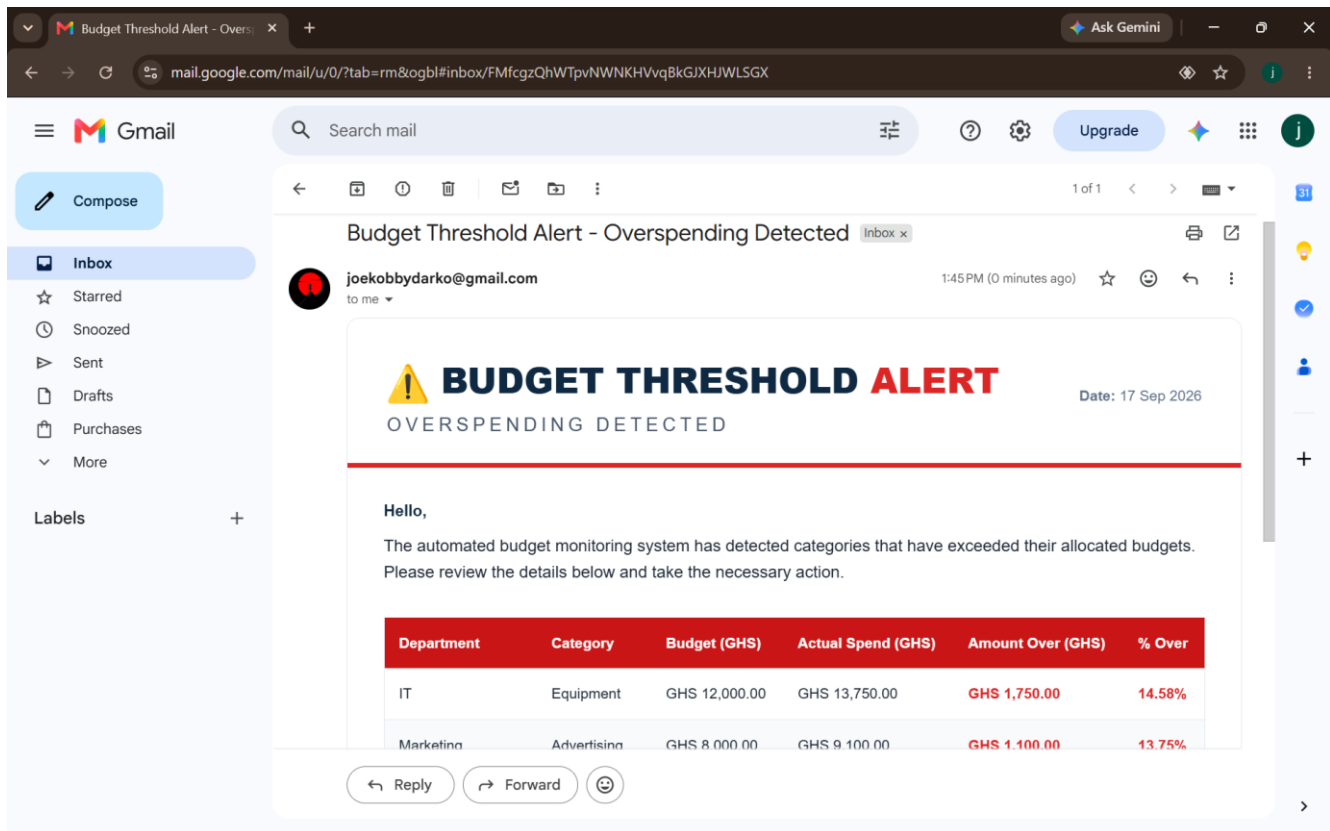


Figure 1. Triggered Budget Threshold Alert email output.

Working Google Apps Script

The following is the exact current script used in the Budget Threshold Alert Tool:

```
function checkBudgetThreshold() {

  const sheet = SpreadsheetApp.getActiveSpreadsheet()
    .getSheetByName("Expense Data");

  const data = sheet.getDataRange().getValues();

  const recipientEmail = "kdarkboadu2005@gmail.com";

  let alerts = [];

  for (let i = 1; i < data.length; i++) {

    const department = data[i][1];
    const category = data[i][2];
    const budget = Number(data[i][4]);
    const actualSpend = Number(data[i][5]);

    if (actualSpend > budget) {

      const amountOver = actualSpend - budget;
      const percentOver = ((amountOver / budget) * 100).toFixed(2);

      alerts.push({
        department,
        category,
        budget,
        actualSpend,
        amountOver,
        percentOver
      });
    }
  }

  MailApp.sendEmail(recipientEmail, "Budget Threshold Alert - Overspending Detected", alerts);
}
```

```

        percentOver
    });
}
}

if (alerts.length > 0) {

    let tableRows = "";

    alerts.forEach(function(alert, index) {

        const background = index % 2 === 0 ? "#ffffff" : "#f8fafc";

        tableRows += `
        <tr style="background:${background};">
            <td style="padding:14px;border-bottom:1px solid #e5e7eb;">
                ${alert.department}
            </td>

            <td style="padding:14px;border-bottom:1px solid #e5e7eb;">
                ${alert.category}
            </td>

            <td style="padding:14px;border-bottom:1px solid #e5e7eb;">
                GHS ${alert.budget.toLocaleString("en-US", {
                    minimumFractionDigits: 2
                })}
            </td>

            <td style="padding:14px;border-bottom:1px solid #e5e7eb;">
                GHS ${alert.actualSpend.toLocaleString("en-US", {
                    minimumFractionDigits: 2
                })}
            </td>

            <td style="
                padding:14px;
                border-bottom:1px solid #e5e7eb;
                color:#dc2626;
                font-weight:bold;">
                GHS ${alert.amountOver.toLocaleString("en-US", {
                    minimumFractionDigits: 2
                })}
            </td>

            <td style="
                padding:14px;
                border-bottom:1px solid #e5e7eb;
                color:#dc2626;
                font-weight:bold;">
                ${alert.percentOver}%
            </td>
        </tr>
        `;
    });

    const today = Utilities.formatDate(
        new Date(),
        Session.getScriptTimeZone(),
        "dd MMM yyyy"
    );

    const htmlBody = `
    <div style="
        font-family:Arial,Helvetica,sans-serif;
        max-width:950px;
        margin:auto;
        background:#ffffff;
        color:#1f2937;
        border:1px solid #e5e7eb;
        border-radius:12px;
        overflow:hidden;">

```

```
<!-- HEADER -->

<div style="
  background:#ffffff;
  padding:30px 35px 20px 35px;
  border-bottom:5px solid #dc2626;">

<table width="100%">
  <tr>

    <td>
      <div style="
        font-size:34px;
        font-weight:800;
        color:#0f2942;">

        ⚠ BUDGET THRESHOLD
        <span style="color:#dc2626;">ALERT</span>

      </div>

      <div style="
        margin-top:5px;
        font-size:18px;
        letter-spacing:4px;
        color:#475569;">

        OVERSPENDING DETECTED

      </div>
    </td>

    <td align="right" style="
      font-size:14px;
      color:#64748b;">

      <strong>Date:</strong> ${today}

    </td>
  </tr>
</table>

</div>

<!-- MESSAGE -->

<div style="padding:30px 35px 10px 35px;">

  <h3 style="
    margin:0 0 10px 0;
    color:#0f2942;">
    Hello,
  </h3>

  <p style="
    font-size:16px;
    line-height:1.6;
    margin:0;">

    The automated budget monitoring system has detected
    categories that have exceeded their allocated budgets.

    <br>

    Please review the details below and take the necessary action.

  </p>

</div>
```

<!-- TABLE -->

```
<div style="
padding:20px 35px;
overflow-x:auto;">

<table width="100%"
cellpadding="0"
cellspacing="0"
style="
border-collapse:collapse;
font-size:14px;
border:1px solid #e5e7eb;">

<thead>

<tr style="
background:#c91515;
color:white;">

<th style="padding:14px;text-align:left;">
Department
</th>

<th style="padding:14px;text-align:left;">
Category
</th>

<th style="padding:14px;text-align:left;">
Budget (GHS)
</th>

<th style="padding:14px;text-align:left;">
Actual Spend (GHS)
</th>

<th style="padding:14px;text-align:left;">
Amount Over (GHS)
</th>

<th style="padding:14px;text-align:left;">
% Over
</th>

</tr>

</thead>

<tbody>

  ${tableRows}

</tbody>

</table>

</div>
```

<!-- ACTION REQUIRED -->

```
<div style="
margin:10px 35px 30px 35px;
padding:20px;
background:#fff1f2;
border-left:6px solid #dc2626;
border-radius:8px;">

<table>
<tr>

<td style="
font-size:30px;
padding-right:15px;">
```

```

        <td>
            <div style="
                color:#b91c1c;
                font-size:18px;
                font-weight:bold;">

                Action Required

            </div>

            <div style="
                margin-top:5px;
                line-height:1.5;">

                Please investigate these overspending categories
                and take corrective action where necessary.

            </div>

        </td>

    </tr>
</table>

</div>

<!-- FOOTER -->

<div style="
    padding:20px 35px;
    background:#f8fafc;
    border-top:1px solid #e5e7eb;">

    <strong style="
        color:#0f2942;
        font-size:16px;">

        Budget Monitoring System

    </strong>

    <br>

    <span style="
        color:#64748b;
        font-size:13px;">

        Automated financial threshold notification.

    </span>

</div>

</div>
`;

GmailApp.sendEmail(
    recipientEmail,
    "Budget Threshold Alert - Overspending Detected",
    "Overspending has been detected. Please view this email in HTML format.",
    {
        htmlBody: htmlBody
    }
);

Logger.log(
    "Professional budget alert email sent successfully."

```

```
);  
} else {  
    logger.log(  
        "All categories are currently within budget."  
    );  
}
```